MS DATA SCIENCE · THESIS DEFENCE

HeatFlood Sentinel

Scalable Big Data and Deep Learning Framework for Heatwave and Flash-Flood Forecasting and Detection in Pakistan

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Roll No. DS-029/2025/26

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Defence date: [PLACEHOLDER — TO BE CONFIRMED]

Internal examiner: [PLACEHOLDER]

External examiner: [PLACEHOLDER]

Thesis results: ERA5, 20 cities, 2015–2022 (test year 2022) · Big Data results: single-node pseudo-distributed Hadoop/YARN/Spark/Kafka cluster

ROADMAP

The talk follows the argument from hazard to evidence to limits

01

Motivation

Deadly heat and floods in Pakistan

02

Problem, RQs & gap

What is missing in 99 reviewed studies

03

Data & methods

ERA5, labels, models, evaluation

04

Big Data architecture

HDFS · YARN · Spark · Kafka

05

Results

Forecasting and detection skill

06

Streaming

Latency, throughput, fault tolerance

07

Limits & conclusion

Limitations, RQ answers, future work

B

Backup

Full tables for questions

01

SECTION

Motivation

Pakistan faces lethal heat and catastrophic floods — often in the same pre-monsoon and monsoon season.

MOTIVATION · HEAT

Karachi 2015: one week of heat killed more than 1,200 people

>1,200

deaths in Karachi during the 17–24 June 2015 heatwave

44.8 °C

maximum temperature on 20 June 2015, 11.1 °C above normal

≈66 °C

heat index on the peak day (high humidity, low wind)

Why it matters for this thesis

The Ministry of Climate Change report calls the event unprecedented in its persistence, not its peak. Multi-day duration is therefore built into the heatwave label — and, as the results show, persistence is also the hardest baseline to beat.

Source: Ministry of Climate Change (MoCC), Technical Report on Karachi Heat Wave, June 2015 (Aug 2015), mocc.gov.pk

MOTIVATION · FLOOD

The 2022 floods affected 33 million people and caused over US$14.9 billion in damage

33 million
people affected

>1,730
lives lost

>$14.9 bn
damages (US$)

≈$15.2 bn
economic losses (US$)

Why it matters for this thesis

2022 was also an exceptionally hot pre-monsoon year. It is the held-out test year here, so every model is judged on an extreme, out-of-distribution season.

Source: Pakistan Floods 2022 Post-Disaster Needs Assessment (PDNA), MoPD&SI with ADB, EU, UN, World Bank, Oct 2022

MOTIVATION · APPROACH

Timely local warnings need data-driven models on scalable data infrastructure

NWP is costly locally

Numerical weather prediction is physically rigorous but expensive at the localised, sub-daily resolution that actionable warnings need.

Climate data is Big Data

ERA5 alone provides hourly global fields at 0.25° over decades: volume, velocity and variety beyond traditional pipelines.

Agencies need explainable alerts

Forecasts and alerts must be accurate, timely and explainable — and run on modest computing infrastructure.

Spark and Kafka make the infrastructure practical; LSTM and ConvLSTM make the models practical. **The literature rarely joins the two — that join is this thesis.**

02

SECTION

Problem, research questions and gap

No reviewed study couples a streaming Big Data pipeline to trained heatwave and flash-flood detectors for Pakistan.

PROBLEM STATEMENT

Five gaps stand between current research and operational warning in Pakistan

i. No reviewed study couples a streaming Big Data ingestion pipeline to a trained heatwave or flash-flood detector.

ii. Processing latency and throughput are rarely reported alongside accuracy.

iii. Validation on Pakistan-specific data — PMD station data in particular — is scarce.

iv. Heatwaves and flash floods are not modelled within one pipeline.

v. Explainability is applied inconsistently.

Need: accurate, timely, explainable alerts produced on modest computing infrastructure.

AIM AND OBJECTIVES

Six objectives carried unchanged from the approved proposal

O1 Focused literature review of Big Data and deep learning for temperature, precipitation, heatwave and flash-flood work, with a South Asian focus.

O2 Identify the data sources, architectures and evaluation practices that most influence accuracy.

O3 Design and implement a distributed pipeline (Spark, Kafka, Hadoop) for ERA5 and PMD data.

O4 Build LSTM and ConvLSTM forecasters plus a heatwave and flash-flood detection module.

O5 Compare against SARIMA and XGBoost baselines, and available NWP outputs.

O6 Assess scalability, throughput and fault tolerance of the streaming pipeline.

RESEARCH QUESTIONS

Four research questions — one answered by review and three by experiment

RQ1	State of the art in Big Data and ML for temperature and precipitation forecasting and heatwave and flash-flood detection, especially in South Asia?	Literature review · Ch. 2
RQ2	Do LSTM and ConvLSTM significantly beat SARIMA and XGBoost on RMSE, MAE and NSE — and how do they compare with NWP — for Pakistan?	Experiments · Ch. 5
RQ3	Can a Spark- and Kafka-based streaming prototype deliver near-real-time detection (latency, throughput, F1, CSI, false alarms)?	Experiments · Ch. 5
RQ4	How do scalability, throughput and fault tolerance vary with data volume and node configuration?	Experiments · Ch. 5

LITERATURE REVIEW · 99 STUDIES

The review of 99 studies maps six gaps to the thesis design

Gap	What the literature lacks	How the thesis responds
G1	Streaming Big Data pipeline coupled to a trained hazard detector	Kafka → Spark stream feeds the detectors
G2	Latency and throughput reported with accuracy	Latency, throughput reported next to CSI/RMSE
G3	Training or validation on PMD station data	PMD adapter built; no data obtained (limitation)
G4	Heatwave and flash flood in one pipeline	Shared pipeline, separate detectors
G5	Global data-driven models benchmarked for Pakistan	Out of scope → future work
G6	Consistent explainability	SHAP feature groups for detectors

Source: Thesis Ch. 2, Table 'Research Gaps and Their Mapping to Thesis Objectives and Research Questions'; 87 original + 12 update-search studies

03

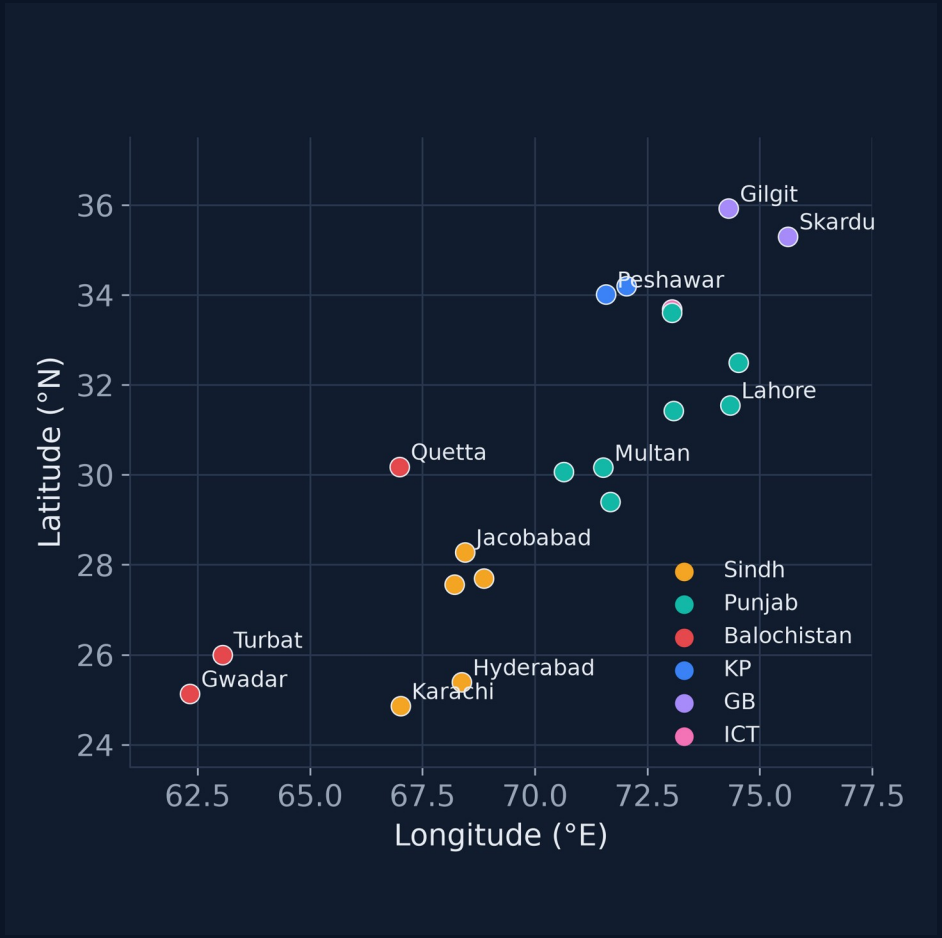
SECTION

Data and methods

ERA5 reanalysis for 20 cities, leakage-safe labels, strong baselines and paired significance tests.

DATA

Thesis results use ERA5 for 20 cities from 2015 to 2022



- **ERA5 reanalysis** via the Open-Meteo archive (models = era5)
- **20 cities**, nearest 0.25° cell, hourly 2015–2022: 1.40 M rows
- **2° regional lattice** (7 × 8 cells) for the ConvLSTM
- **Split:** train 2015–2020 · validation 2021 · test 2022 (held out)
- **Extension to 1990–2026 in progress:** 1990–1999 + 2015–2022 already in HDFS (3.11 M rows); not used for thesis numbers
- **PMD stations:** adapter implemented, no data obtained

Source: DEVIATIONS.md D4–D5, D8; results/bigdata/hdfs_load.json; thesis Appendix C (study cities)

HAZARD DEFINITIONS

Labels are percentile-based and fitted on training years only

Label	Definition	Positives (train / val / test)	Used for
Heatwave (relative)	Daily Tmax > local calendar-day p95 (±7-day window) on ≥3 consecutive days	593 / 261 / 551 city-days	Detection target
Heatwave (absolute)	Daily Tmax ≥ 40 °C on ≥3 consecutive days	4,596 / 620 / 870	Reported
Flash-flood proxy A	6-h rain ≥ max(50 mm, local p99) and antecedent precipitation index > p75	17 / 1 / 2 windows	Too rare: descriptive only
Flash-flood proxy B	Trailing 24-h rain ≥ local p99.5 of daily totals	929 / 79 / 405 windows	Detection target

Flood labels are rainfall-extreme proxies, not observed floods — ERA5 does not observe floods.

Source: Thesis Ch. 3 hazard-definition table; results/bigdata/spark_etl_parity_2015_2022.json (label prevalence); DEVIATIONS.md D1

MODELS AND EVALUATION

Every model faces persistence and climatology on identical test samples

Baselines

- Persistence (incl. diurnal)
- Climatology (train-only)
- SARIMA
- XGBoost (default hyper-parameters)
- Rule-based persistence detector

Deep learning

- Daily and hourly LSTM
- ConvLSTM encoder–forecaster on the 2° lattice (60,198 parameters, 6 h ahead)
- CNN grid heatwave detector
- Calibrated XGBoost detectors with SHAP

Evaluation

- RMSE, MAE, NSE; skill vs persistence and climatology
- CSI, F1, POD, FAR; ROC-AUC, PR-AUC, Brier
- Diebold–Mariano and block bootstrap, Holm-corrected
- Thresholds tuned on 2021, tested on 2022

Source: Thesis Ch. 3 (models, evaluation design); Appendix C hyper-parameters; DEVIATIONS.md D7

04

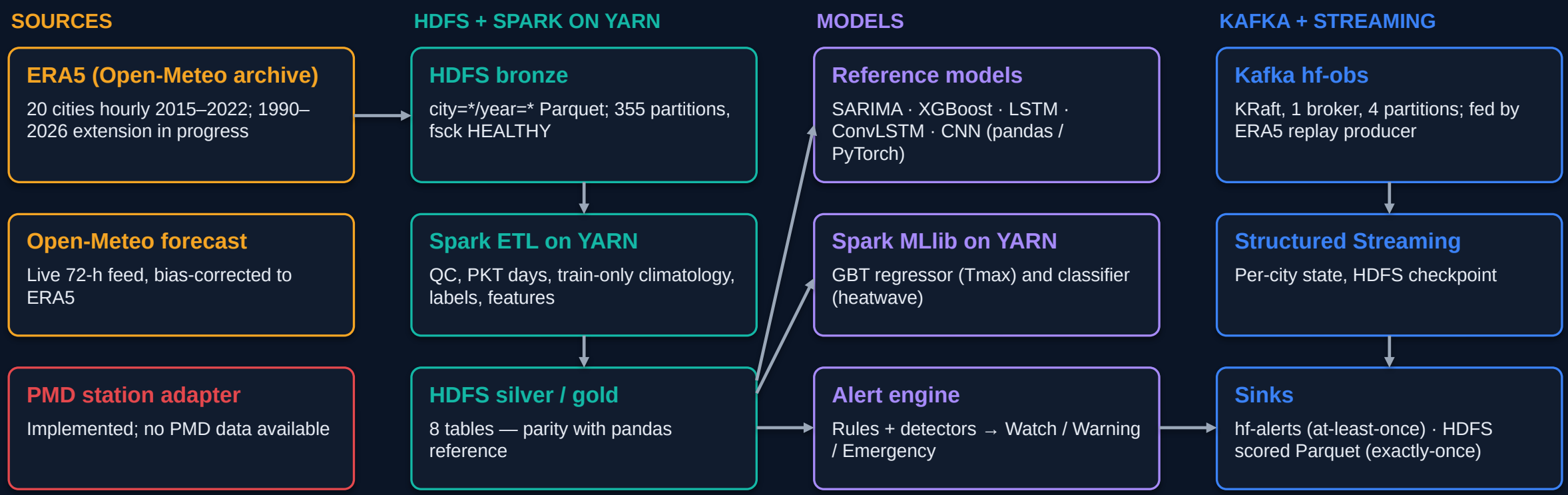
SECTION

Big Data architecture

One code path from HDFS to alert — executed on a single-node pseudo-distributed Hadoop, YARN, Spark and Kafka cluster.

ARCHITECTURE

Batch and streaming paths share storage, labels and alert logic



All daemons on one 2-vCPU / 7.7 GB node — a single-node pseudo-distributed cluster. Real services and protocols; no multi-node parallelism.

Source: heatflood-sentinel docs/bigdata.md; DEVIATIONS.md D3, D8; thesis Ch. 4 architecture

CLUSTER SETUP

A real Hadoop stack ran end-to-end on one machine

Component	Version and configuration
HDFS	Hadoop 3.3.6 · NameNode + DataNode · replication 1
YARN	1 ResourceManager + 1 NodeManager (4 GB / 4 vcores advertised)
Spark	3.5.9 on YARN, client mode · 1 executor × 2 cores, 1 GB heap
Kafka	3.9.2 KRaft, 1 broker · hf-obs 4 partitions, hf-alerts 1
Host	2 vCPU aarch64 · 7.7 GB RAM · OpenJDK 17 · Python 3.12
Evidence	9 YARN applications logged · 16 Big Data tests passed

What it demonstrates

- Real HDFS blocks, YARN containers, Kafka log
- Spark shuffle, state store, checkpoints

What it does not

- No multi-node parallelism or network cost
- No fault tolerance across machines

Source: results/bigdata/RUN_LOG.md (application IDs, versions, pytest); results/bigdata/streaming_kafka_spark.json (environment); DEVIATIONS.md D8

SPARK ETL PARITY

Spark ETL on YARN reproduces the pandas pipeline on all 8 tables

Table (2015–2022)	Rows (pandas = Spark)	Result
Silver · city hourly	1,402,560	**PASS**
Silver · city daily	58,420	**PASS**
Gold · climatology hourly T2m	175,680	**PASS**
Gold · climatology hourly precip.	175,680	**PASS**
Gold · climatology daily Tmax	7,320	**PASS**
Gold · flash-flood 6-h labels	233,760	**PASS**
Gold · daily features	58,420	**PASS**
Gold · hourly features	1,402,560	**PASS**

8 / 8

tables match: same keys, labels exact

1.7×10^{-4}

largest float difference (heat index, float32)

241.8 s

Spark ETL wall time on YARN (1.40 M hourly rows)

Source: results/bigdata/parity_2015_2022.json (atol 1e-3, rtol 1e-4); spark_etl_parity_2015_2022.json; DEVIATIONS.md D8

05

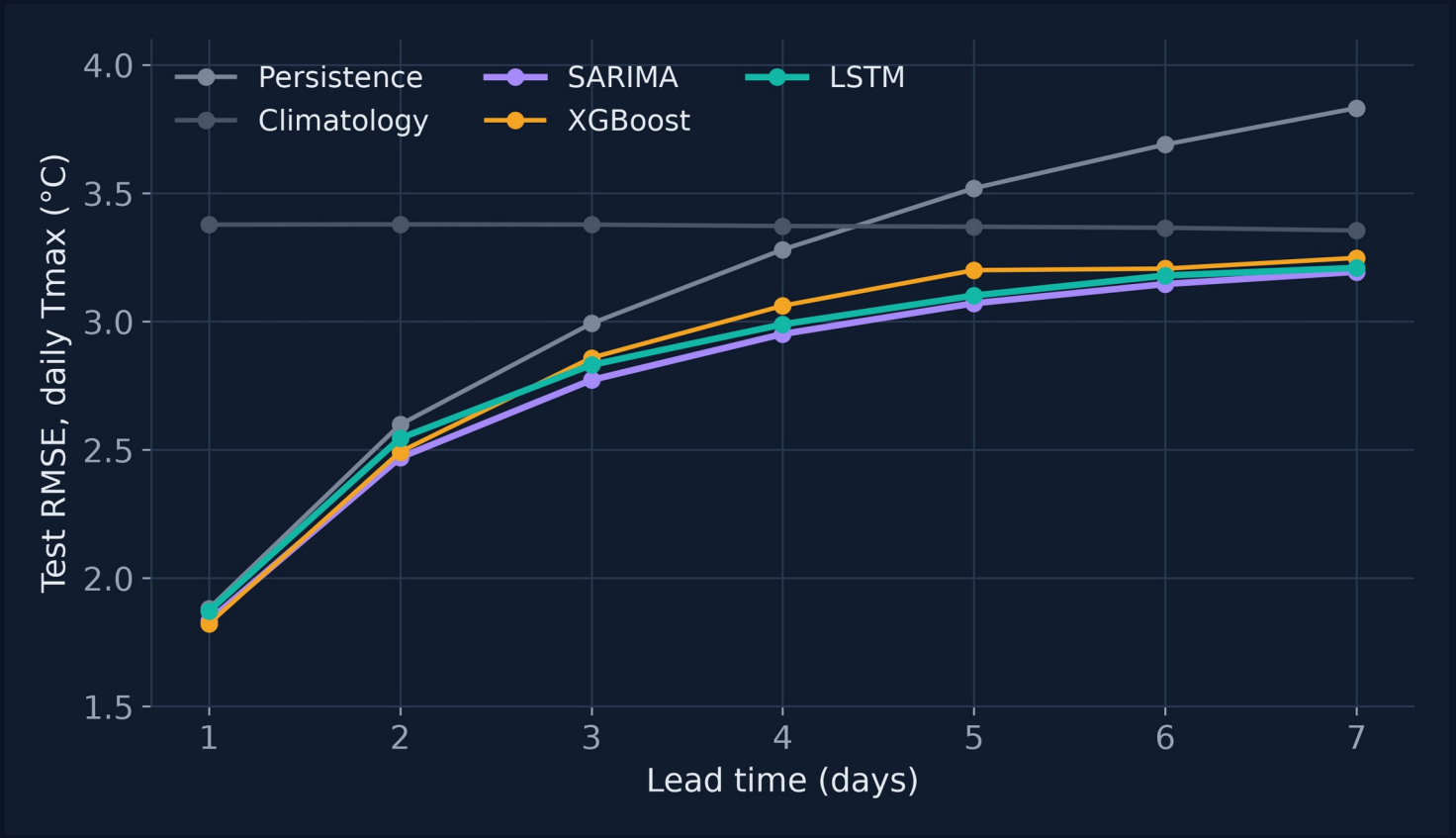
SECTION

Results

Strong baselines are hard to beat. Deep learning wins for hourly and gridded temperature — not for daily Tmax or next-day heatwaves.

RESULTS · DAILY TMAX (RQ2)

Daily Tmax: SARIMA matches or beats the LSTM at every lead

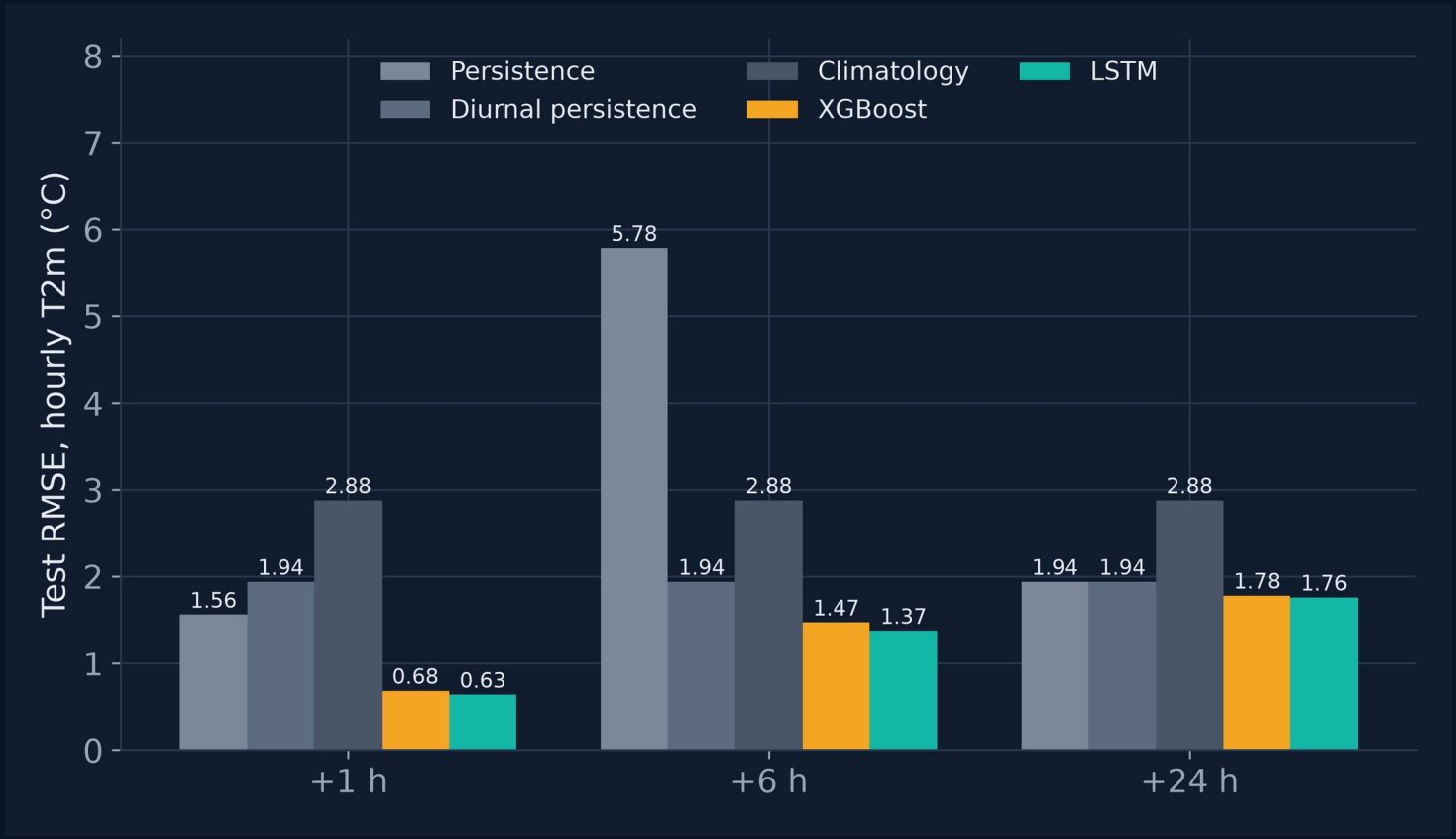


- **+1 day RMSE:** XGBoost 1.822, SARIMA 1.834, LSTM 1.871, persistence 1.881 °C
- **LSTM beats persistence significantly from +3 days** (Holm-corrected DM test)
- **Precipitation:** LSTM gives small significant gains at 3–7 days; no model forecasts heavy daily rain usefully

Source: results/tables/leaderboard_daily.csv, significance_daily.csv; RESULTS.md (test 2022)

RESULTS · HOURLY T2M (RQ2)

Hourly LSTM beats XGBoost and persistence for temperature at every lead



0.634 °C

LSTM RMSE at +1 h (XGBoost 0.679, persistence 1.558)

1.372 °C

LSTM RMSE at +6 h (XGBoost 1.469, persistence 5.783)

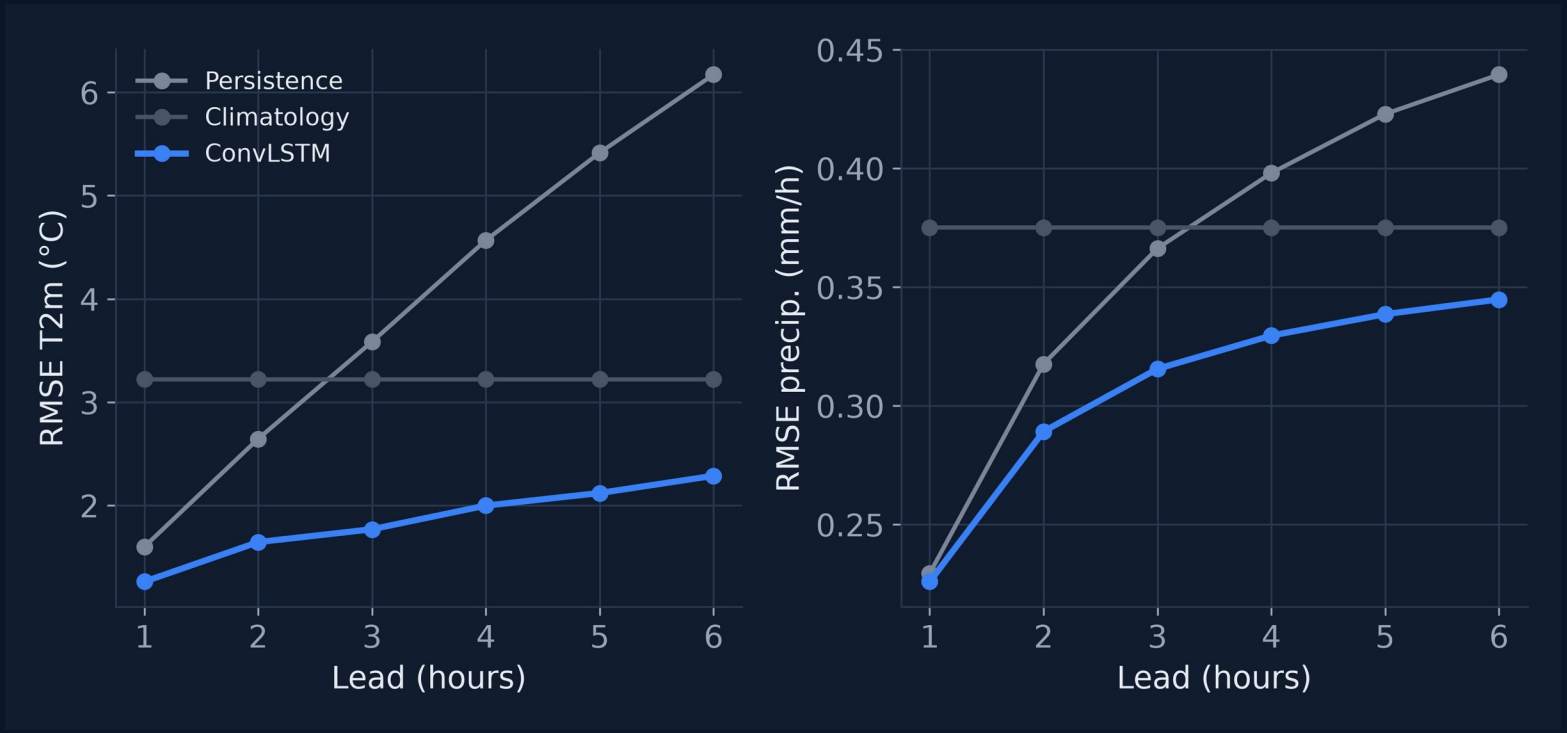
1.756 °C

LSTM RMSE at +24 h (XGBoost 1.779, persistence 1.938)

Source: results/tables/leaderboard_hourly.csv; RESULTS.md; thesis Ch. 6 RQ2 answer

RESULTS · GRIDDED CONVLSTM (RQ2)

ConvLSTM beats persistence and climatology at every lead on the 2° lattice

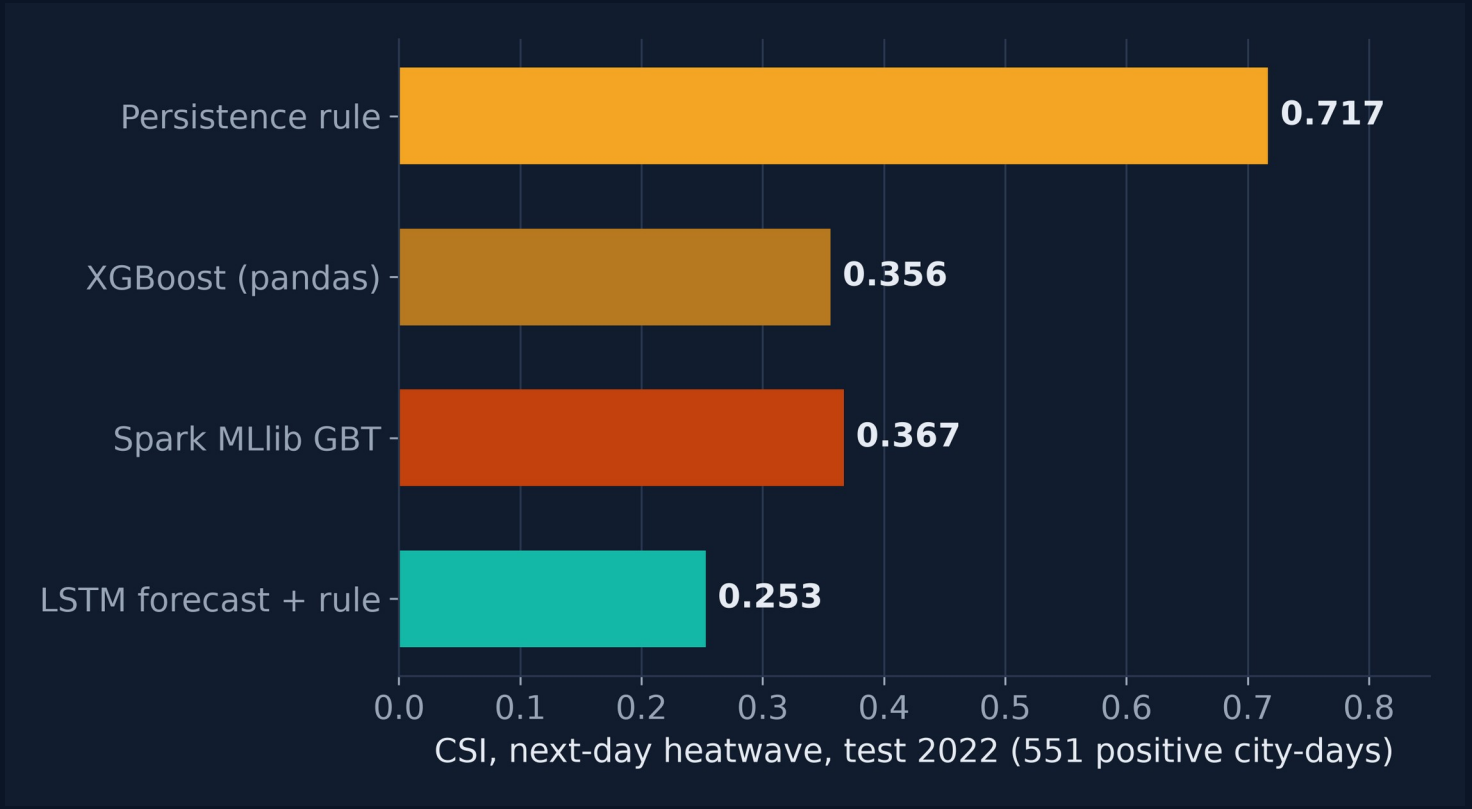


- **T2m +1 h:** 1.262 vs persistence 1.600 °C
- **T2m +6 h:** 2.287 vs persistence 6.171 °C
- **Precip. +1 h:** 0.226 vs 0.229 mm/h — marginal
- **Caveat:** CPU prototype, 2° grid, ~6 epochs, one seed

Source: results/tables/grid_convlstm_test.csv (490,224 samples per lead); DEVIATIONS.md D2

RESULTS · HEATWAVE DETECTION (RQ3)

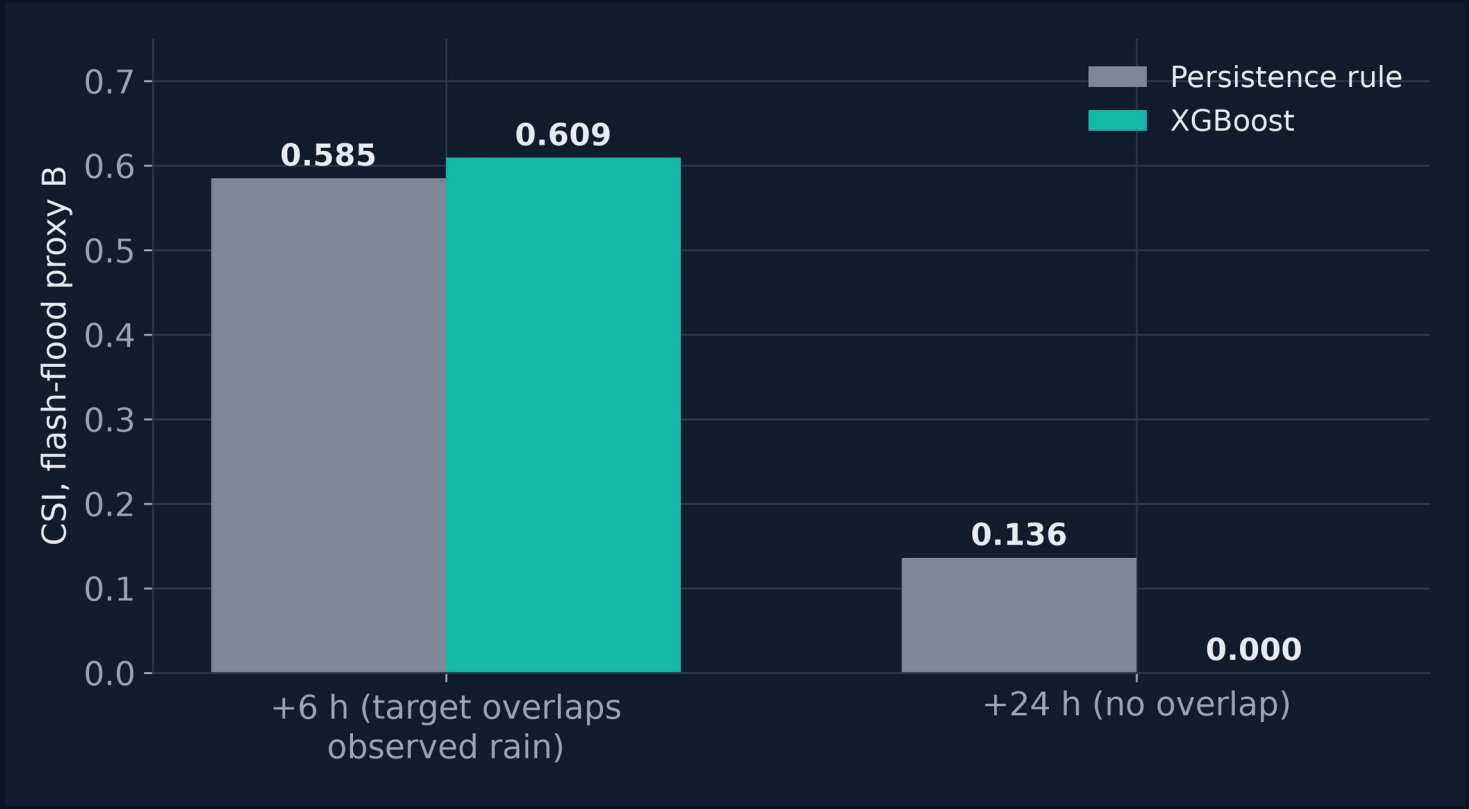
Persistence detects next-day heatwaves twice as well as XGBoost



- **Persistence CSI 0.717** vs XGBoost 0.356 (ROC-AUC 0.943)
- **Spark MLlib GBT:** CSI 0.367 — on par with XGBoost
- **+3 days:** XGBoost issues no alerts at thresholds tuned on milder 2021 (CSI 0)
- **Lesson:** heatwaves persist — the baseline sets the bar

Source: results/tables/detection_test.csv; results/bigdata/mllib_baseline.json (test 2022, 551 positives of 7,280)

The flood proxy +6 h CSI of 0.609 is an upper bound and not a forecast

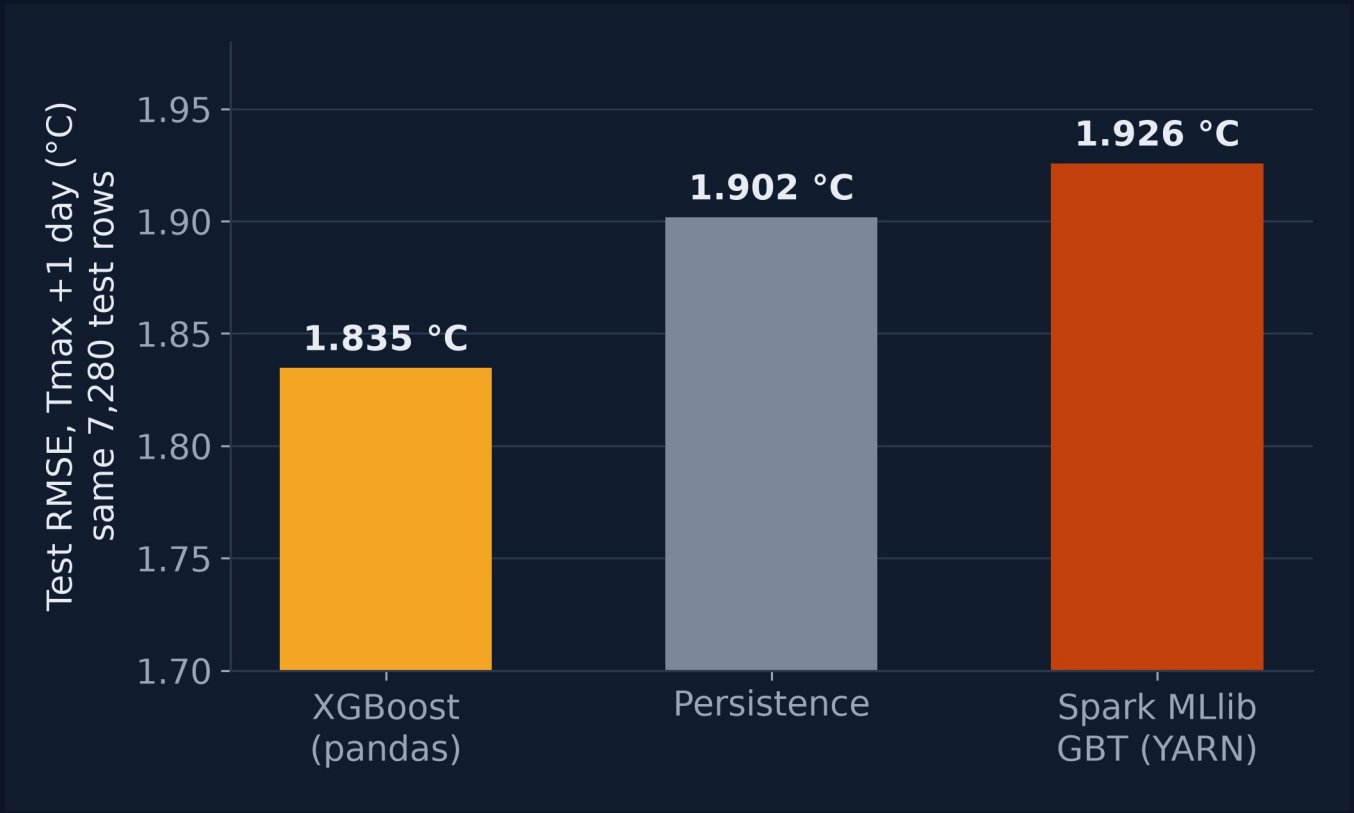


- **+6 h:** XGBoost CSI 0.609 vs persistence 0.585; PR-AUC 0.814
- **Window overlap:** target shares 18 of 24 h with rain already observed
- **+24 h (honest forecast):** CSI 0 at tuned threshold; ROC-AUC 0.939
- **Labels are rainfall proxies,** not observed floods

Source: results/tables/detection_test.csv (405 positive windows in 2022); DEVIATIONS.md D1

RESULTS · SPARK MLLIB ON YARN

Spark MLlib matches XGBoost for heatwave detection but not for Tmax regression



1.926 °C

MLlib GBT Tmax +1 d RMSE: XGBoost 1.835, persistence 1.902

CSI 0.367

MLlib GBT heatwave +1 d (XGBoost 0.356, persistence 0.717)

AUC 0.947

ROC-AUC; PR-AUC 0.618; threshold 0.15 tuned on 2021

Source: results/bigdata/mllib_baseline.json (YARN application_1792965347508_0004, seed 42; same 7,280 test rows)

06

SECTION

Streaming

Streaming is fast enough for warnings and lost nothing under failure — on one node.

STREAMING · LOCAL RUNTIME (THESIS RESULTS)

The local stream processed a six-month season with zero lost alerts

87,840

events replayed: Apr–Sep 2022, 20 cities, hourly

210.5 /s

events per second sustained (183 daily batches)

2.37 s

p95 to process one day of data for all 20 cities

6 fault modes · 0 crashes

duplicate, corrupt, out-of-order, delayed, missing, outage; 0 duplicate alerts

0 lost · 0 duplicate

alerts after kill-and-restart (restored from checkpoint)

Source: results/streaming_local.json (runtime: local fallback, in-process queue + StreamProcessor); thesis RQ3 answer

STREAMING · KAFKA → SPARK ON YARN

Kafka to Spark processed 2,739 events per second with a 5.5 s p95 to HDFS



2,739 /s

processing rate on a 58,560-event backlog (3 micro-batches)

5.5 s

p95 event → HDFS Parquet at 100 events/s (p50 2.8 s)

3.2 s

p95 event → Kafka hf-alerts (54 alerts)

Source: results/bigdata/streaming_kafka_spark.json (YARN apps _0005, _0006); single-node pseudo-distributed cluster

STREAMING · FAULT TOLERANCE

Killing the Spark driver lost and duplicated nothing in the HDFS sink

21,600 / 21,600
Kafka events = HDFS sink rows (distinct offsets)

0 · 0
missing offsets · duplicate rows in Parquet sink

212 = 212
alerts identical to offline reference; features max |diff| 0.0

0 s	45 s	+43 s	+34.9 s
250 events/s	SIGKILL driver	restart from HDFS checkpoint	drain backlog

Kafka alert topic is at-least-once: 254 messages, 42 replayed from the in-flight batch — consumers must de-duplicate by event id.

Source: results/bigdata/streaming_kafka_spark.json → kill_restart (YARN apps _0007 → _0008)

SCALABILITY (RQ4)

Batch feature cost scales linearly with volume on one node



- **E1 volume:** linear, $R^2 = 0.9999$ over a 4× increase
- **E2 workers:** speed-up 1.88 with 2 processes (efficiency 0.94)
- **Not shown:** scaling across nodes — needs a multi-node cluster

Source: results/scalability_local.json (2-vCPU node, process-level); thesis RQ4 answer

07

SECTION

Limitations, conclusions and future work

What the evidence supports, what it does not, and what comes next.

LIMITATIONS

Six limitations bound every claim in this thesis

PMD station data unavailable

All results use ERA5 reanalysis; the PMD adapter was tested only on synthetic files.

Proxy flood labels

Rainfall-extreme proxies, not observed floods; +6 h target overlaps observed rain.

Single seed for some analyses

ConvLSTM, XGBoost and MLlib use one seed; only the daily LSTM has three seeds.

One exceptional test year

2022 only; percentile thresholds from six training years (2015–2020).

Single-node pseudo-distributed cluster

No multi-node scaling or cross-machine fault tolerance measured.

No NWP comparison

No archived 2022 NWP forecasts; short CPU training of neural networks.

Source: Thesis Ch. 5 threats to validity and Ch. 6 limitations; DEVIATIONS.md D1–D8; thesis Table 'Daily LSTM over three random seeds'

ANSWERS TO THE RESEARCH QUESTIONS

Each research question gets a qualified answer backed by evidence

RQ	Answer	Key evidence
RQ1	Strong baselines + LSTM/ConvLSTM; streaming Big Data rarely coupled to detectors	99-study review; gaps G1–G6
RQ2	Partly: yes for hourly and gridded temperature; no for daily Tmax	LSTM 0.634 vs XGB 0.679 °C (+1 h); SARIMA ≥ LSTM daily
RQ3	Partly: timely and robust; skill limited and hazard-dependent	Persistence CSI 0.717 > XGB 0.356; flood +6 h 0.609 (upper bound)
RQ4	Yes on a single node; not yet shown on a multi-node cluster	R^2 0.9999; 2,739 ev/s; 21,600/21,600 after kill

Source: Thesis Ch. 6 'Answers to the Research Questions'; results/.json; results/bigdata/*.json*

CONTRIBUTIONS

Five contributions that others can reuse and verify

1. An end-to-end open-source pipeline coupling streaming ingestion to trained heatwave and flash-flood-risk detectors (G1, G4).
2. Latency, throughput and fault tolerance reported next to RMSE and CSI (G2).
3. An honest benchmark against persistence, climatology, SARIMA and XGBoost with Holm-corrected significance tests.
4. Spark ETL parity (8/8 tables) and exactly-once HDFS sink verified on a single-node Hadoop/YARN/Spark/Kafka stack.
5. Public, reproducible artefacts: tagged GitHub release with CI and tests, plus a live site refreshed every 6 hours.

Source: Thesis Ch. 6 contributions; github.com/SirAseef/heatflood-sentinel (release v0.1.0); results/bigdata/RUN_LOG.md

FUTURE WORK

Next steps close the gaps this thesis could not

Ground truth

Validate against PMD stations; replace proxies with NDMA/PDMA flood-event records.

1990–2026 record

Finish the download, rebuild the lake on HDFS and retrain on the full period.

Extreme-aware learning

Quantile or extreme-value losses; recalibrate seasonal, POD-constrained thresholds.

NWP and global models

Benchmark archived NWP, GraphCast and Pangu-Weather for Pakistan (G5).

Multi-node cluster

Run multi-broker Kafka and multi-worker Spark; test node-failure tolerance.

More architectures and hazards

Attention and graph models; drought and cold waves.

Source: Thesis Ch. 6 future work; DEVIATIONS.md D8 (data extension status)

CONCLUSION

I request approval of the thesis as a reproducible and honestly bounded early-warning baseline for Pakistan

- Deep learning helps for hourly and gridded temperature
- Persistence remains the bar for next-day heatwaves
- Streaming is timely and lossless on a single-node pseudo-distributed cluster
- Next: PMD ground truth, 1990–2026 record, multi-node cluster



Live site

siraseef.github.io/heatflood-sentinel



Code repository

github.com/SirAseef/heatflood-sentinel

Defence date: [PLACEHOLDER]

Examiners: [PLACEHOLDER]

Thank you — I welcome the committee's questions.

B

SECTION

Backup slides

Full tables for questions: forecasting leaderboards, detection metrics and fault-injection results.

Daily Tmax and precipitation RMSE by model and lead in 2022

Model	Tmax +1 d	Tmax +3 d	Tmax +7 d	Precip. +1 d	Precip. +3 d	Precip. +7 d
Persistence	1.881	2.992	3.832	6.346	7.566	7.642
Climatology	3.377	3.378	3.355	5.841	**5.840**	**5.652**
SARIMA	1.834	**2.772**	**3.194**	**5.656**	5.996	5.809
XGBoost	**1.822**	2.859	3.248	5.697	5.994	5.827
LSTM	1.871	2.831	3.209	5.677	5.946	5.756

RMSE in °C (Tmax) and mm (precipitation); best per column in green. Test year 2022, 20 cities.

Source: results/tables/leaderboard_daily.csv; RESULTS.md

Full detection metrics for both hazards on the 2022 test year

Hazard · lead	Model	CSI	POD	FAR	F1	ROC-AUC	PR-AUC
Heatwave · 1d	Persistence rule	0.717	0.835	0.165	0.835	—	—
Heatwave · 1d	LSTM forecast + rule	0.253	0.258	0.066	0.404	—	—
Heatwave · 1d	XGBoost	0.356	0.407	0.261	0.525	0.943	0.589
Heatwave · 3d	Persistence rule	0.348	0.517	0.485	0.516	—	—
Heatwave · 3d	LSTM forecast + rule	0.007	0.007	0.000	0.014	—	—
Heatwave · 3d	XGBoost	0.000	0.000	—	0.000	0.814	0.209
Flash flood · 6h	Persistence rule	0.585	0.738	0.262	0.738	—	—
Flash flood · 6h	XGBoost	0.609	0.723	0.206	0.757	0.984	0.814
Flash flood · 24h	Persistence rule	0.136	0.240	0.760	0.240	—	—
Flash flood · 24h	XGBoost	0.000	0.000	—	0.000	0.939	0.216
Heatwave · 1d	Spark MLlib GBT	0.367	0.443	0.317	0.537	0.947	0.618

Source: results/tables/detection_test.csv; results/bigdata/mllib_baseline.json; FAR '—' = no alerts issued

Six injected fault types caused no crash and no duplicate alerts

Fault mode	Messages in	Dead-letter (reason)	Alerts vs baseline	Crashed	Duplicate alerts
Duplicate messages	15,213	333 duplicate	384 (+0)	No	0
Corrupt records	14,880	333 schema/range	383 (-1)	No	0
Out-of-order	14,880	0	384 (+0)	No	0
Delayed beyond watermark	14,846	299 late beyond watermark	383 (-1)	No	0
Missing data	14,880	0	383 (-1)	No	0
Source outage	13,440	0	342 (-42)	No	0

Baseline window: 384 alerts. Kill-and-restart: killed after batch 18, restored from batch 15, 0 lost and 0 duplicate alerts.

Source: results/streaming_local.json → chaos (31-batch window, local runtime)